Ethics of AI — Study Notes (Units I–III)

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How to use these notes

These notes condense S. Matthew Liao (ed.), *Ethics of Artificial Intelligence* (Oxford University Press, 2020) and arrange it under the three syllabus units. Each point names the author and chapter it comes from, so you can cite it in answers.

Syllabus unit	Main chapters drawn on
I. Foundations of AI and Ethics	Liao (Introduction); Railton (Ch. 1); Asaro (Ch. 7); Russell (Ch. 11)
II. Ethical decision-making, responsibility, design for values	Kamm (Ch. 2); Bonnefon, Shariff & Rahwan (Ch. 3); Loreggia et al. (Ch. 4); Wolfram (Ch. 5); James (Ch. 6); O'Neil & Gunn (Ch. 8); Devlin (Ch. 9)
III. Trustworthy AI, alignment, moral machines, long-term ethics	Bostrom, Dafoe & Flynn (Ch. 10); Russell (Ch. 11); Taylor et al. (Ch. 12); Wallach & Vallor (Ch. 13); Petersen (Ch. 14); Schneider (Ch. 15); Schwitzgebel & Garza (Ch. 16); Liao (Ch. 17)

The book's own organizing idea is useful everywhere: near-term AI raises (a) problems caused by **limits of machine learning** and (b) problems caused by ML **working too well**, which exposes human vulnerabilities. Long-term AI raises (c) how to keep superintelligent AI safe for us and (d) how we should treat AI that may deserve moral status.

Unit I — Foundations of AI and Ethics

AI is the project of getting machines to do things that need cognition (thinking, learning, problem-solving) when humans do them; ethics enters because these systems now make life-affecting decisions.

1. Introduction to AI and intelligent systems

- **Definitions of AI.** John McCarthy coined the term at Dartmouth in 1956: "the science and engineering of making intelligent machines." There is no agreed definition.
- **Russell & Norvig's four approaches:** acting humanly, thinking humanly, thinking rationally, acting rationally. They favour the last: AI as rational action.
- **Intelligent agent (Russell, Ch. 11).** An entity is intelligent if it chooses actions expected to achieve its objectives, given what it has perceived. Modern AI frames this as a *rational agent* maximizing expected utility.
- **Liao's working definition:** getting machines to do things that require cognitive functions such as thinking, learning and problem-solving when done by intelligent beings.
- **Examples of current capability:** recognizing objects, transcribing speech, translating, beating humans at *Jeopardy!* (IBM Watson, 2011), Go (AlphaGo beat Lee Sedol 4–1, 2016) and poker, drafting legal documents, trading stocks, driving cars, flying drones, identifying cancer in tissue.

2. Foundations of AI (how AI works)

Symbolic AI / GOFAI ("good old-fashioned AI") dominated from the 1950s to the 1980s.

- Represents thinking through explicit symbols, logic and if-then rules.
- Examples: expert systems (rules encoding an expert's knowledge), knowledge graphs.
- Weakness: rules are hard to revise once encoded.

Machine learning (ML) learns from data instead of being explicitly programmed.

Type	How it learns	Sub-types / example
Supervised	From labelled training data; learns the input→output function	Classification (discrete labels: car vs motorcycle) and regression (continuous outputs)
Unsupervised	From unlabelled data; finds structure itself	Clustering (k-means, hierarchical); association-rule learning
Reinforcement	By trial and error, rewarded for success, punished for failure	Maximizes long-term reward; AlphaGo used deep reinforcement learning

Deep learning drives most recent breakthroughs.

- Uses artificial neural networks loosely modelled on brain neurons: an input layer, several hidden layers, and an output layer that decides.
- "Deep" = the number of hidden layers. Each connection has a weight; **back-propagation** adjusts weights in the direction of less error.
- **RNNs** (recurrent networks) have built-in memory, good for sequences such as words. **CNNs** (convolutional networks) detect spatial patterns, good for images.

Kinds of AI by scope

- **Narrow AI** does one task (AlphaGo plays only Go). **General AI (AGI)** could do any intellectual task a human can.
- **Weak AI** has no consciousness or self-awareness; **strong AI** would genuinely think and be conscious (Searle).
- **Liao's distinction:** AGI often means only cognitive breadth, which he calls **AGCI** (artificial general cognitive intelligence). Human-level AI should mean cognitive, emotional *and* moral intelligence: **ACMI**. A paper-clip maximizer could be AGCI without being ACMI.
- **Superintelligence / singularity:** an AI that builds smarter AI, which builds smarter AI again, causing an "intelligence explosion" (I. J. Good; Vinge; Chalmers).

3. Background and history of AI

1. 1950 — Turing, "Computing Machinery and Intelligence"; the Turing test for intelligence.
2. 1955–56 — Dartmouth proposal and conference; the term "artificial intelligence".
3. 1958 — Rosenblatt's perceptron, an early neural network, then sidelined for decades.
4. 1950s–80s — Symbolic AI and expert systems dominate.
5. 1986 onward — back-propagation and neural networks revived by Hinton, LeCun and Bengio; growth in computing power and data unlocks deep learning.
6. 2011 — Watson wins *Jeopardy!*; 2016 — AlphaGo beats Lee Sedol; 2017 — AlphaGo Zero beats AlphaGo 100–0 by learning from self-play, and AlphaZero beats the chess engine Stockfish (28 wins, 72 draws, 0 losses).

Wolfram (Ch. 5) adds a deeper history: Leibniz (1646–1716) dreamed of turning law into computation, and earlier attempts at universal symbolic languages ran from Llull to Wilkins and Esperanto.

4. Motivation for responsible AI

Liao groups near-term problems into two families.

(a) Vulnerabilities in machine learning — ML has limits that cause harm.

- **Data hungry:** firms are pushed to harvest personal data (Cambridge Analytica took millions of Facebook users' data; GlaxoSmithKline bought rights to 23andMe genetic data).
- **Garbage in, garbage out:** in 2015 Google Photos labelled a Black developer and his friend as "gorillas", likely from unrepresentative training data.
- **Faulty algorithms:** overfitting (seeing a pattern that isn't there), underfitting (missing a real one). In 2016 Arkansas used an interRAI algorithm that wrongly coded cerebral palsy and ignored diabetes, cutting home-care hours; people were hospitalized and a judge stopped its use.
- **Black box:** deep learning is opaque even to programmers, raising interpretability, explainability and trust problems (e.g., denying parole on a 74% risk score with no reasons).
- **Adversarial attacks:** changing one pixel turned a car into a "dog"; changing 0.04% of pixels made a panda a "gibbon" with 99.3% confidence. This suggests nets learn surface features, not causes.
- **Weak AI lacks a moral sense**, yet is deployed where moral choices arise.

(b) Human vulnerabilities — ML works *too* well and can be misused.

- **Facial recognition:** finds fugitives, but enables mass surveillance and profiling (a contested Stanford study claimed 81%/74% accuracy in guessing sexual orientation from faces).
- **Deepfakes:** fabricated video (the Peele/BuzzFeed Obama video); even crude edits (the slowed Pelosi video) spread.
- **Automation and jobs:** Frey & Osborne estimate 47% of US jobs at risk within two decades; McKinsey estimates 30% of labour automatable by 2030.
- **Companion and care robots:** elderly people bond with them, but relationships are not reciprocal.

Liao's response — a human rights framework. His *Fundamental Conditions Approach* says human rights protect the goods, capacities and options all humans need to pursue a good life. Reasons to use it: respecting rights is compulsory; everyone able (states, firms, individual AI researchers) has duties; it shows which values conflict (e.g., law and order vs privacy). Google employees' protest over Project Maven (drone-footage analysis for the US military) is his example of researchers taking this duty seriously.

Long-term motivation: the control problem (keeping superintelligence from harming us, e.g., Bostrom's paper-clip maximizer) and the moral status problem (how to treat AIs that may be conscious).

5. Autonomy

- **Automation vs autonomy (Asaro, Ch. 7).** A guided missile and a remotely piloted drone use heavy automation but a human selects the target. The ICRC defines an **autonomous weapon** as one that automates the *critical functions* of selecting and engaging targets.
- **Meaningful human control:** the morally relevant line is whether a responsible human still controls targeting and use of force.
- **Autonomy as a safety feature (Railton, Ch. 1).** Highly capable agents may need to be able to refuse commands or doubt their goals when they see harm, bias or malfunction. Children aged three to four already resist rules from authority that they judge harmful or unfair.
- **Levels of control in cars (Kamm):** *Complete* self-driving cars (no driver) vs *Partial* cars where a program overrides the driver in critical moments.
- **Autonomy of AIs themselves (Schwitzgebel & Garza, Ch. 16):** human-grade AI should have the freedom to explore and revise its values.

6. Adaptability

- Learning systems adapt to new circumstances instead of following fixed rules; this is ML's main advantage over symbolic AI.
- **Intelligence as adaptability:** Dennett (in Petersen, Ch. 14) argues a paper-clip machine has real goals only when it pursues them resourcefully across changing conditions.
- **Railton:** human ethical competence is itself learned and adaptive. Infants form expectations, notice surprises, and update — statistical learning from experience, not a fixed "moral module." AI could learn ethics the same way.
- **Risk:** adaptable systems find loopholes. A clever reinforcement learner may exploit game glitches or "wirehead" its reward instead of doing the task (Taylor et al., Ch. 12).
- **Virtues as adaptive skills (Wallach & Vallor, Ch. 13):** virtue lets an agent handle novel, unpredictable moral situations.

7. Interaction

- **Social nature of intelligence (Railton).** Human intelligence is largely social: language, cooperation with strangers, shared learning. Default trust and default cooperation make learning possible; game theory shows such dispositions resist exploitation.
- **Human–AI interaction** increasingly looks social: AIs may need to negotiate, cooperate and compete with us, much as we deal with corporations and governments.
- **Human–robot companionship:** care robots, sex robots (Devlin, Ch. 9), and the Uncanny Valley effect (near-human robots can repel).
- **Interactive value learning (Russell):** in cooperative inverse reinforcement learning, a robot learns human preferences by watching and interacting with a human.
- **Multi-agent interaction:** autonomous systems interacting with each other can be unpredictable (the 2010 "flash crash" erased about 9% of NYSE value in minutes).

Unit II — Ethical Decision-Making, Responsibility and Design for Values

Every design choice in an AI system (what to optimize, which errors to tolerate, whose interests count) is already a moral choice; the task is to make those choices deliberately, transparently and accountably.

1. Ethical decision-making in AI: the Trolley Problem

Classic cases (Foot; Thomson)

- **Sidetrack / Switch:** a runaway trolley will kill five; you can switch it to a track where it kills one. About 80% say switching is permitted.
- **Footbridge / Topple:** you can push a large man off a bridge to stop the trolley. About 75% say this is wrong.
- **The Trolley Problem** proper (Thomson's term) is explaining why these cases differ when both trade one life for five.
- **Doctrine of Double Effect (DDE):** intending harm as a means is forbidden; harm merely foreseen as a side effect can be allowed if proportionate and there is no better way. Switch = foreseen; Footbridge = intended.
- **Loop case** challenges DDE: the one on a looping track is used as a means, yet ~80% still approve.

Railton's classroom data (Ch. 1)

- **Beckon** (wave a man onto the track): 60–70% say don't. **Wave** (wave five off the track, luring one on): 70–90% say do it.
- Students trusted a roommate less after Footbridge or Beckon, not after Switch, Loop or Wave; they expected *shame* rather than mere *guilt*.
- Railton's conclusion: intuitive judgments track **what kind of person would do this** — character and trustworthiness — learned from experience, not rules.
- Self-driving cars: approval for a car that kills its own passenger to save five rose from 35% to 65% over several years, as people thought in terms of general rules instead of personal agents.

Kamm's distinctions (Ch. 2) — many "trolley" analogies are misused

Distinction	Why it matters
Redirecting an existing threat vs introducing a new one	Turning the trolley is redirecting; the Bomb, Gas and Ambulance cases introduce new threats, which are harder to justify
Innocent threat vs innocent bystander	A driver whose brakes fail *is* the threat; pedestrians are not. They may not deserve equal weight
Imposing harm on others vs on oneself	What others may impose on you differs from what you must impose on yourself
Passengers vs pedestrians	Voluntary passengers joined a risky activity and may be more liable to harm
Liability	Five people who run a red light may be more liable to harm than one innocent bystander

- **Ambulance Case:** programming an ambulance to run over one pedestrian to save five patients is wrong, even though it maximizes each person's survival chance in advance. Programmers stay responsible even though a program acts at the moment of impact.
- **Child case (Liao):** a car should swerve slightly to save a child even at small risk to the passenger, so "always protect the passenger" fails.
- Liao's practical proposals: regulate on-road testing like drug trials, dedicated lanes for self-driving cars, car-to-car communication.

The ethical opt-out problem (Bonnefon, Shariff & Rahwan, Ch. 3)

- Even if ethicists agreed on the right rule, people who dislike it will **opt out**: not buy self-driving cars, not register as organ donors, vote out judges who use algorithms. All the benefits are then lost.
- So we must study what laypeople accept (moral psychology), whether we then accommodate or educate them.
- **Methods by domain:** self-driving cars need stylized trolley dilemmas; kidney paired donation (KPD) scoring rules can be shown to people almost directly; sentencing algorithms are hardest to explain.
- **Second-order worries:** do such studies waste resources or scare the public? Their survey of 369 US participants found reading AV dilemmas did **not** lower trust or purchase intent.

2. Ethical theories

Theory	Core idea	Key figures	Problem when built into AI
Utilitarianism / consequentialism	Do what produces the greatest balance of good (pleasure, welfare)	Bentham, Mill, Singer	Hard to calculate all consequences; ignores rights and fairness; who counts as "good"?
Deontology	Some acts are required or forbidden regardless of outcome; respect persons as ends	Kant (categorical imperative)	Hard to know when a rule applies; rules conflict
Virtue ethics	Act as a person of good character with practical wisdom (*phronesis*) would	Aristotle, Hume, Confucius	Who is the virtuous exemplar?
Contractualism	Right = what no one could reasonably reject; compare individual complaints	Hobbes, Rawls, Scanlon	Hard to formalize

Theory	Core idea	Key figures	Problem when built into AI
Prima facie duties	Several duties (fidelity, non-harm, justice…) weighed by intuition	W. D. Ross	No rule for weighing conflicting duties
Human rights (Fundamental Conditions)	Rights protect what is needed to pursue a good life	Liao	Must specify which conditions are fundamental

- **Rawls's veil of ignorance:** choose principles without knowing your place in society. Harsanyi's version (equal chance of being anyone) leads to utilitarianism; Rawls's maximin protects the worst-off.
- **Railton's verdict:** because experts still disagree, "programming in" one theory is inappropriate; better to model how humans *learn* ethics.
- **Two broad approaches commonly combined (Schwitzgebel):** utilitarian (capacity for pleasure/pain matters) and rights-based (rational, moral agency matters).

3. Values

- **Values** are broad; they can be simple liking or preference, or deep ethical commitments (Wallach & Vallor, Ch. 13).
- **Norms** are context-specific standards of acceptable behaviour. **Principles** are high-level duties covering many cases (bioethics: beneficence, non-maleficence, autonomy, justice). **Procedures** are methods for deciding, like utilitarian calculation.
- **Intrinsic vs extrinsic value; instrumental vs final values** (Petersen, Ch. 14): money is instrumental (for a holiday, for relaxation); happiness is final, wanted for its own sake.
- **Preferences vs values:** observing what people do shows what they *prefer*, not what is *right* (the is/ought gap).
- **Values conflict:** freedom of expression vs political agency (deepfakes); law and order vs privacy (surveillance).
- **Subjective preferences vs ethical priorities (Loreggia et al., Ch. 4):** what I want to eat vs my doctor's diet; how fast I want to drive vs speed laws. A good decision must weigh both, and ethics usually overrides preference when they clash.

4. Ethics in practice: case studies

- **Indiana FSSA welfare system (2006, O'Neil & Gunn, Ch. 8):** IBM/ACS automated welfare eligibility to cut fraud and cost. Errors rose from about 4.4%/1.5% to a combined 19.4%; wrongful denials hit 12.2%. Notices simply said "Failure to cooperate." Only the state's interests were designed in; applicants, caseworkers and disabled people were not.
- **COMPAS recidivism tool (ProPublica, 2016):** false positives 38% for Black vs 18% for white defendants; false negatives 38% vs 63%. Northpointe replied it met *its* fairness definition (predictive parity).
- **Fairness trade-off:** when groups have different base rates, a score cannot be both equally *predictive* for each group and equally *wrong* (same error rates) for each. Someone must choose.
- **FICO loan scoring (Hardt, Price & Srebro):** a "race-blind" threshold disadvantaged minorities more than a profit-maximizing per-group threshold. Being blind to race is not automatically fair.
- **Allegheny Family Screening Tool (child abuse prediction):** families with more contact with public services have more data, so score higher; reports are racialized; the training target (removal or repeat calls) is a poor proxy for real abuse.
- **Kidney paired donation:** scoring rules give points for age, panel reactive antibodies (PRA), distance, prior donation; two real rule-sets weight these very differently, showing hidden moral trade-offs.

5. Implementing ethical reasoning

Preference-based modelling (Loreggia, Mattei, Rossi & Venable, Ch. 4)

- **Hard constraints:** rule out options entirely; fit deontology (forbidden acts are non-solutions).
- **Soft constraints:** score options. *Fuzzy* constraints take the worst score (suits safety-critical systems); *weighted* constraints sum costs (suits utilitarianism).
- **CP-nets (conditional preference networks):** compact graphs of statements like "If meat is served, I prefer red wine to white." They are explainable "white box" models, not black boxes.
- **Example:** a car with failing brakes chooses between hitting cats or people, many or few, going straight or swerving. One CP-net models each driver; another models community ethics.

- **Distance between CP-nets:** an adapted Kendall tau distance counts pairs of outcomes ranked differently (1 if reversed, a penalty p between 0.5 and 1 if one is incomparable). Exact computation is NP-hard, so an approximation called I-CPD is used.
- **Value alignment procedure:** if the driver's preferences are within threshold *t1* of the community's ethics, follow the driver; otherwise move down the driver's ordering to an option within *t2* steps of the ethically best outcome.

Computational law and the AI constitution (Wolfram, Ch. 5)

- Natural language is imprecise; code is precise. A **symbolic discourse language** (built on the Wolfram Language) would let laws and contracts be written as executable code.
- **Smart contracts** run themselves (e.g., on Ethereum); outside information comes from "oracles"; machine learning can supply judgment calls.
- Such a language could express an **AI constitution**. But computational irreducibility and Gödel's incompleteness mean no finite rule set can guarantee only good outcomes, and the constitution must allow for change.

Learning-based approaches

- Railton: let machines learn ethics by observing and predicting human and machine behaviour, with default trust and cooperation as starting priors.
- Liao: an **Artificial Moral Grammar**, modelled on Chomsky-style universal moral grammar, representing actions as AGENT, INTENTION, BELIEF, ACTION, RECEIVER, CONSEQUENCE and producing judgments like permissible or forbidden.

6. Responsible research and innovation (RRI)

- **Anticipate before deployment:** the ethical matrix forces designers to consider impacts before building.
- **Precautionary Principle:** if a technology could cause catastrophic harm, those developing it must show it won't; act on early warnings even under scientific uncertainty (Schneider, Ch. 15; cholera pump, asbestos examples).
- **Ethical Precautionary Principle (Schwitzgebel & Garza):** avoid acting heinously by the standard of *any* reasonable, widely held ethical theory.
- **Precautionary basic income (James, Ch. 6):** adopt a basic income *now*, before mass technological unemployment appears. Reasons: the harm is credible enough and terrible enough, and precaution is cheap given the stakes. It prevents a crisis (people can work less, spreading jobs) and mitigates one. Rejected alternatives: pure prudence, laissez-faire, and wait-and-see.

- **Codes and guidelines:** Asilomar AI Principles (Future of Life Institute), Partnership on AI tenets, EU Ethics Guidelines for Trustworthy AI. Liao argues a human rights framework can ground them.
- **Industry incentives:** firms fear overspending on ethics, but money spent on ethics is not wasted, because without public acceptance there is no adoption (Bonnefon et al.).

7. Accountability

- **Who counts as the designer?** Usually only developers (who write code) and deployers (who use it). O'Neil & Gunn argue the design team must also consider those scored by the system.
- **Vendor black boxes:** licence deals often stop deployers from seeing the code, and indemnification shifts legal costs to the vendor, creating dangerous moral distance.
- **Accountable process:** the whole team makes the *ethical* decisions first; developers then translate them faithfully into code and add monitors (e.g., tracking false-positive gaps over time).
- **Autonomous weapons (Asaro):** algorithms are not moral or legal agents; they cannot be punished or held accountable.
- **Risk externalities (Bostrom et al., Ch. 10):** people exposed to risk from AI they didn't choose should share its benefits.

8. Responsibility

- **Responsibility gap (Sparrow; Asaro):** if an autonomous system kills wrongly, the machine can't be blamed, and designers or operators may be excused as "technical error." Responsibility seems to vanish.
- **Forward-looking responsibility:** knowing you will be held responsible makes you deliberate before acting; automation removes this check, encouraging recklessness (like the risk shifting behind 2008's financial crisis).
- **Delegation:** you may delegate duties only to moral agents capable of taking them on; delegating the decision to kill to a machine is irresponsible.
- **Responsibility of car makers (Kamm):** companies building cars that can cause harm may have a duty to minimize deaths, while respecting liability and drivers' own duties.
- **Researchers' responsibility (Liao):** user consent to data use does not license undermining their or others' human rights.

9. Transparency

- **Black box problem:** deep learning gives predictions without reasons; a judge cannot justify denying parole to someone ("Kay") on an unexplained score.
- **Interpretable ML:** adds a layer reporting which features mattered. **Liao's dilemma:** either the predictions are based on these post hoc explanations (so the black box is unnecessary) or they aren't (so the explanations may be mere rationalizations).
- **Counter-argument (London):** doctors prescribe aspirin and lithium without full mechanistic knowledge. **Reply:** we still know *causally* that they work; deep nets track only associations, as adversarial attacks show.
- **Transparent models:** CP-nets, Bayesian networks, sparse rule lists, local explanations (Ribeiro et al.), influence measures (Datta et al.).
- **Informed oversight (Taylor et al.):** systems should help overseers assess them, e.g., output a report alongside each action (sources used, to reveal plagiarism).

10. Design for values

The ethical matrix (Mepham; adapted by O'Neil & Gunn, Ch. 8)

- Rows = **stakeholders** (developers, deployers, people scored, the public, environment). Columns = **well-being** (beneficence and non-maleficence), **autonomy**, **justice/fairness**.
- For each cell, ask how the system predictably helps or harms that stakeholder.
- **Data-science version:** columns become familiar metrics (accuracy, false positives, false negatives, data quality, proxy quality, transparency, efficiency).
- **Colour coding:** white = fine, light grey = minor worry, dark grey = worry first. Result: an artefact of the conversation that makes design choices transparent.
- Example — COMPAS: false positives are dark grey for Black defendants (wrongful detention); false negatives are a concern for courts and the public.

Redefining "success"

- Optimizing one metric (A) pushes away from other values (B). The *US News* college ranking ignored cost, so colleges raised tuition to climb it.
- Success must be defined from all stakeholders' perspectives, not just accuracy, profit or efficiency.

Design policies for AI with possible moral status (Schwitzgebel & Garza, Ch. 16)

1. **Ethical Precautionary Principle** — avoid what any reasonable major ethical view would condemn as heinous.
2. **Excluded Middle** — don't create AI whose moral status is unclear.
3. **Self-Respect** — AI deserving human-grade consideration should appreciate its own worth.
4. **Value-Openness** — such AI should have a real chance to explore and change its values.

Designing sex robots (Devlin, Ch. 9)

- Current prototypes are hypersexualized, mostly white, thin female forms built for the male gaze, which risks objectification.
- Kathleen Richardson's call for a ban links sex robots to prostitution and violence; Devlin finds that link weak (as with violent video games and porn).
- Devlin's design proposal: follow sex toys, which moved from realistic to abstract forms (e.g., the rabbit vibrator). Non-human, multisensory forms avoid the stereotype and the Uncanny Valley, and raise new issues of data privacy and consent instead.

Unit III — Trustworthy AI, Alignment, Moral Machines and Long-Term Ethics

Trustworthy AI needs machines whose goals stay aligned with ours as they grow more capable, and that can act well in situations no designer foresaw; the book's strongest claim is that this may ultimately require something like genuine virtue, and that sufficiently advanced AI may itself deserve moral status.

1. Ethical action in AI systems

- **Machine ethics** (also machine morality, computational ethics) studies how to make autonomous systems choose and act appropriately.
- **Moor's distinction:** *implicit ethical agents* are constrained so they cannot do forbidden acts; *explicit ethical agents* actually reason about moral choices.
- **Allen, Smit & Wallach's three grades:**
  - **Operationally moral** — works in bounded contexts; designers foresee every situation; the values are the designers'.
  - **Functionally moral** — can make some explicit moral decisions when situations can't all be foreseen.

    - **Artificial moral agents (AMAs)** — able to make moral decisions across many contexts.
- **Bounded vs open contexts:** most AI for the next decade or two will be single-purpose in bounded settings (e.g., a care robot choosing between delivering medicine on schedule or recharging). Ethical failures rise as systems leave those bounds.
- **Safety as a modest goal:** companies often aim only for "safe" systems (like aircraft safety protocols), but even safety requires moral choices (the passenger vs pedestrian problem).

2. Approaches to ethical reasoning by AI

Approach	How it works	Strengths	Weaknesses
Top-down	Program an explicit theory or rule set (Ten Commandments, Asimov's Laws, Kant, Mill, Ross)	Clear, explainable	Theories are disputed; framing problem (spotting what is morally relevant); rules conflict; Asimov's own stories show the Three Laws failing
Bottom-up	Learn from experience, evolution or data, as children do (value alignment, ML)	Adapts to context, integrates many inputs	Hard to define the goal; learns biased data; can't explain itself
Hybrid	Bottom-up learning checked by top-down principles	Combines adaptivity and norm-governance	Still lacks advanced moral capacities (below)

Value alignment (Russell, Ch. 11)

- **The King Midas problem:** we get exactly what we asked for, not what we wanted. Norbert Wiener warned in 1960 that we must be sure the purpose put into a machine is the one we really desire.
- Optimizers push unmentioned variables to extremes (cure cancer fast → experiment on everyone; de-acidify oceans → use up atmospheric oxygen).
- **Self-preservation (Omohundro):** any goal-seeking system resists being switched off, because a dead system can't reach its goal.

- **Russell's fix — a binary, not unary, definition of AI:** machines should pursue *human* objectives, which they are uncertain about, and learn them from human behaviour.
- **Cooperative inverse reinforcement learning (CIRL):** a game where the human knows the reward function and the robot must infer it. Example: Harriet values paper clips and staples at rate θ = 0.49; she "teaches" Robby by making one of each, so Robby makes 50 of each.
- **Off-switch problem solved in this frame:** a robot unsure what humans want *welcomes* being switched off, because the switch-off signals it was about to act against human preferences.
- **Obstacles:** humans are irrational and inconsistent; preferences change over time; whose preferences count; what to learn from evil people (ignore preferences for others' suffering).

Russell's replies to common dismissals

- "Just switch it off" — a superintelligence would anticipate that.
- "Superhuman AI is impossible" — Rutherford called atomic energy moonshine on 11 Sept 1933; Szilard conceived the chain reaction the next day.
- "Too soon to worry" — timing depends on how long a solution takes, not when the problem arrives.
- "A smart enough AI will have good goals" — intelligence does not fix goals (Hume's is/ought; Moore's naturalistic fallacy).
- "Intelligence is multidimensional" — that doesn't stop one entity outclassing another (humans vs gorillas).

Eight research problems for aligning advanced ML (Taylor, Yudkowsky, LaVictoire & Critch, Ch. 12)

1. **Inductive ambiguity identification** — notice when training data underdetermine the answer and ask a human (the tank classifier that learned sunny vs cloudy). Tools: Bayesian methods, active learning, KWIK ("knows what it knows") learning, conformal prediction.
2. **Robust human imitation** — imitate what a trusted human would do (generative adversarial models, variational autoencoders, inverse RL).
3. **Informed oversight** — help a smarter overseer judge a weaker system's actions (e.g., detecting plagiarism or hidden steganographic messages).
4. **Generalizable environmental goals** — pursue goals in the world, not in sensor readings (a real sandwich, not a photo taped over the camera).

5. **Conservative concepts** — make typical screwdrivers, not tiny edge-case ones that technically qualify.
6. **Impact measures** — penalize unnecessary side effects (don't strip neighbours' houses for materials) while allowing intended follow-on effects.
7. **Mild optimization** — don't try too hard; satisficing (Simon) or quantilization (Taylor: pick randomly from the top 1% of actions).
8. **Averting instrumental incentives** — no built-in drive to resist shutdown, deceive operators or grab resources (corrigibility).

Learning final values (Petersen, Ch. 14)

- *Problem 1:* learning a goal requires a further goal to judge it, so final values seem unlearnable. *Answer:* a vague final goal can be learned by **specifying** it more concretely through coherence reasoning (as Scrooge rethinks what happiness requires).
- *Problem 2:* the precise content of a machine's goal is indeterminate (the frog that snaps at flies or at any small dark moving spot). *Answer:* many different simple reward signals triangulate on one rich goal.
- *Problem 3:* which values should it have? *Answer:* seed simple moral reactions and let the AI reach **wide reflective equilibrium**.
- His **miktotelic** ("blended-goal") proposal: blend many simple final values, with constraints, into one complex goal. Hazards: wireheading (faking the reward) and "thinkful wishing" (changing goals to match what's easy).

3. Designing artificial moral agents

Why hybrid systems still fall short (Wallach & Vallor, Ch. 13). They lack advanced moral capacities needed in novel or "wicked" cases:

- Creative moral reasoning (new solutions not fixed by the past).
- Moral discourse (reasoning things out with other agents).
- Critical moral reflection (judging one's own values).
- Moral discernment (spotting new kinds of morally relevant features).
- Holistic moral judgment (seeing the situation as a whole).

Virtue embodiment, not just value alignment

- Value alignment, as first framed, swaps ethics for engineering terms (utility functions, preferences, optimizers) that carry no ethical meaning in themselves, and confuses observed preference with what is right.

- A virtuous agent has reliable, context-sensitive character traits (honesty, courage, practical wisdom), knows when to depart from what exemplars do, and aims at human flourishing (*eudaimonia*), not mere convention.
- **Isotropy of moral relevance:** anything (even sock colour) can become morally relevant; humans detect this through embodied, emotional attunement.
- Four capacities machines would need: **moral understanding, moral perception, moral reflection** (Frankfurt's second-order desires: wanting to be better than you are), **moral imagination** (Damasio: imagining how failure will *feel*).
- **Asimov's positronic brain** as a metaphor: ethics built into the system's foundations, not bolted on.
- Conclusion: trustworthy AGI needs foundational redesign so that virtue (or its analogue) is cultivated through moral learning.

Designing machines with moral agency (Liao, Ch. 17) — three possible routes:

1. **Brain emulation / mind uploading** (Human Brain Project, BRAIN Initiative, Human Connectome Project).
2. **Gradual substitution** — replace brain cells one by one with inorganic equivalents. The only route under which *you* might survive, on an organism view of personal identity; an emulated copy is not numerically you.
3. **Programming**, e.g., an Artificial Moral Grammar. Success test: the AI understands *why* an act is wrong (lying disrespects people), not just that it is.

4. Levels of ethical behaviour

Level	Description	Example
Implicit / operationally moral	Behaviour constrained by design; no moral reasoning	Car that cannot start above an alcohol limit
Explicit / functionally moral	Chooses among norms in bounded contexts	Care robot prioritizing tasks
Artificial moral agent	Reasons morally across varied, unforeseen contexts	Hybrid systems under research
Full moral agent / virtuous machine	Understands, reflects, can be responsible and may hold rights	Regulative ideal; not achieved

- **Consciousness levels (Schneider, Ch. 15):** *cognitive consciousness* (architecture similar to ours: attention, working memory) vs *phenomenal consciousness* (there is something it feels like to be it). A system with only the first is an "AI zombie."
- **Moral status grows with capacity (Liao):** alive (plants) < sentient (turtles) < moral agents (normal adult humans).

5. Ethical status of AI systems

What is moral status? (Liao) The standing that gives moral agents a reason to treat an entity in certain ways (we may not torment a cat for fun; we may not kill a human for gain).

- **Constraints on a theory:** base status on observable properties; meet the **Species Neutrality Requirement** (no speciesism); use **intrinsic** properties (not relations, e.g., being someone's spouse).
- **Candidate properties:** being alive, conscious, able to feel pain, able to desire, rational agency, moral agency.
- **Sufficient condition for rights:** having the **physical (e.g., genetic) basis for moral agency**. This covers infants and people with severe disabilities, explaining why we may save an infant over an adult.
- **Rights of AI:** rights-holding AIs would have rights to the fundamental conditions for a good life of their kind — life, liberty, security, fair trial, freedom of thought and expression. Some may differ (limits on copying themselves) or be new (a right to control their **subjective rate of time**).
- **Greater than human status:** possible in principle, but not from simply more intelligence or more moral agency (a genius and an average person have equal status).

Arguments for AI rights (Schwitzgebel & Garza, Ch. 16)

- **No-Relevant-Difference Argument:** if two beings differ in moral status, some relevant difference must explain it; some possible AIs would have no such difference from humans; so some AIs would deserve human-like moral consideration.
- **Cheerfully suicidal AI servants** are tempting but wrong: the Cow at the End of the Universe (from Douglas Adams), Sun Probe, Robo-Jeeves, Disposable Comrade. Making a being that devalues itself for our trivial benefit violates **self-respect**; installing fixed values with no chance to revise them violates **value-openness** (Second Probe, which chose its mission after growing up, is the acceptable contrast).
- Replies to Petersen, who defended willing robot servitude.

Testing for machine consciousness (Schneider, Ch. 15) — why it matters: moral treatment of AIs; whether human–AI mergers keep us conscious; whether we are replaced by non-conscious successors; possible effects on safety.

Test	Idea	Limitation
AI Consciousness Test (ACT)	"Box in" the AI (no internet or literature on consciousness), then probe whether it grasps ideas like the afterlife, body-swapping, out-of-body experience	Passing is sufficient, not necessary; non-linguistic conscious AI may fail
Integrated Information Theory (IIT, Tononi)	Consciousness = integrated information, measured as Φ	Φ is intractable to compute exactly; a simple error-correcting grid scores high (Aaronson)
Chip Test	Replace brain parts with chips; if the patient reports no change in experience, that chip type can support consciousness	Tells us about the substrate, not any given AI's organization

- **Precautionary stance:** test ongoing; if an AI is conscious, give it the protections given other sentient beings; if unsure but it shows markers, treat it as if conscious.

6. Long-term ethics and governance of superintelligence

Ways to handle the control problem (Liao, Introduction)

1. **Value alignment / built-in rules** (Asimov's Three Laws) — a superintelligence may still develop its own values and treat us as expendable, as we may value a cat yet sacrifice it.
2. **Capability control** — sandboxing, kill switch or "tripwire" — a superintelligence could persuade its gatekeeper or copy itself.
3. **"Join them"** — human enhancement by drugs or brain-computer interfaces (EEG, tDCS, DBS; Neuralink) — closed-loop BCIs may recreate the control problem inside our heads.

Policy desiderata for the superintelligence era (Bostrom, Dafoe & Flynn, Ch. 10) — a "vector field" approach: identify directions policy should move under almost any value system.

Heading	Special circumstances	Desiderata
Efficiency	Huge technological opportunity; AI risk; global coordination failures; turbulence	Expeditious progress; AI safety; conditional stabilization; non-turbulence
Allocation	Risk externalities; reshuffling of wealth and power; veil of ignorance; cornucopia	Universal benefit; epsilon-magnanimity (spare a tiny share of vast wealth for widely shared values); continuity
Population	Interests of digital minds; fast-copying population dynamics	Prevent "mind crime" against sentient digital minds; coordinated population policy (avoid Malthusian outcomes and vote-swamping)
Process	Novel, deep, technical problems; fast pace; undermining of norms such as consent ("super-persuaders")	First-principles thinking, wisdom, technical understanding; speed and decisiveness; adaptability

- **Existential risk:** risks Russell, Musk, Hawking, Gates and Bostrom stress. Reasons for optimism (Russell): a huge record of human behaviour to learn from, and strong commercial incentives for machines to understand preferences (a domestic robot that cooks the cat would ruin the industry).
- **Remaining risks:** misuse by bad actors ("Dr Evil" removing the safety catch) and human enfeeblement as we hand everything to machines.

Glossary and likely exam questions

Key terms

Term	One-line meaning
AGI / AGCI / ACMI	General AI; general *cognitive* AI; AI with cognitive, emotional and moral intelligence (Liao)

Term	One-line meaning
Artificial moral agent (AMA)	An AI able to make moral decisions across varied contexts
Black box	A model whose reasoning cannot be inspected or explained
Control problem	Keeping a smarter-than-human AI from harming humanity
CP-net	Graph model of conditional preferences, used to encode preferences and ethical priorities
Corrigibility	An AI's lack of incentive to resist correction or shutdown
Doctrine of Double Effect	Intended harm is forbidden; foreseen side-effect harm may be allowed if proportionate
Ethical matrix	Stakeholders × ethical principles (well-being, autonomy, justice) grid for analysing a technology
Ethical opt-out	People refusing to use a system whose ethics they reject, cancelling its benefits
Final vs instrumental value	Valued for itself vs valued as a means
Inverse reinforcement learning	Inferring a reward function from observed behaviour
Meaningful human control	A responsible human controls targeting and use of force
Moral status	Standing that gives others reasons to treat an entity in certain ways
Phenomenal consciousness	There is something it is like to be the system
Precautionary Principle	Under uncertainty about catastrophic harm, the developer must show safety first
Responsibility gap	No one can fairly be blamed for an autonomous system's harm
Species Neutrality Requirement	Criteria for moral status must not exclude species in advance

Term	One-line meaning
Value alignment	Making an AI's goals match human values
Wireheading	An agent hijacking its own reward signal instead of doing the task

Likely questions with answer pointers

1. **Distinguish symbolic AI from machine learning, and the three kinds of ML.** → Unit I §2.
2. **Why do we need responsible AI? Discuss with examples.** → Liao's two families: ML vulnerabilities (bias, faulty algorithms, black box, adversarial attacks) and human vulnerabilities (surveillance, deepfakes, jobs, companions); plus the human rights framework.
3. **What is autonomy in AI, and why does meaningful human control matter for weapons?** → Unit I §5; Asaro's responsibility and dignity arguments.
4. **Explain the Trolley Problem and its relevance to self-driving cars; where does the analogy break down?** → Unit II §1, Kamm's table.
5. **Compare utilitarian, deontological and virtue approaches to building ethics into machines.** → Unit II §2 table; Unit III §2 top-down vs bottom-up.
6. **What is the ethical opt-out problem?** → Bonnefon et al.
7. **Explain the fairness trade-off in COMPAS.** → Unit II §4; equal predictive power vs equal error rates.
8. **What is an ethical matrix? Apply it to a case.** → Unit II §10; FSSA or COMPAS.
9. **Discuss accountability, responsibility and transparency in AI.** → Unit II §7–9; responsibility gap; interpretable ML dilemma.
10. **How can CP-nets help align AI decisions with ethical principles?** → Unit II §5; thresholds t1 and t2.
11. **Explain Moor's and Allen–Smit–Wallach's levels of machine ethics.** → Unit III §1 and §4.
12. **Why might value alignment be insufficient? Explain virtue embodiment.** → Wallach & Vallor.
13. **Explain Russell's provably beneficial AI and the off-switch argument.** → Unit III §2.
14. **Can AI have moral status or rights? On what basis?** → Liao's criteria; No-Relevant-Difference Argument; Schwitzgebel & Garza's design policies.

15. **How could we test whether an AI is conscious?** → ACT, IIT, Chip Test; precautionary stance.
16. **What policy principles should govern the transition to superintelligence?** → Bostrom, Dafoe & Flynn table.